Original Article

Association between smartphone glances during application-based navigation and pedestrian navigation performance: a real-world experimental study

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Background: The percentage of people owning smartphones is rapidly increasing due to the digitalization of modern society. Among the smartphone's features, route guidance using navigation applications establishes spontaneous and independent outdoor mobility, and can potentially support people's health. Despite some known concerns regarding smartphone use while walking, navigation performance and eye-tracking behavior while walking with navigation applications in real-world environments are unclear. This study aimed to investigate the association between gaze behavior, particularly glancing at smartphones, and navigation performance among adult users of navigation applications while walking.

Methods: This cross-sectional experimental study involved 33 healthy adults aged 20 years and older (20–81 years, 66.7% female) not requiring assistance with walking or activities of daily living. A route navigation task (RNT) of approximately 1,300 m was performed in a real outdoor environment. The participants were required to complete the route as quickly as possible, following the instructions of the navigation application. Simultaneously, the fixation time, number of fixations, number of saccades, total saccade angle, and number of glances at the smartphone were recorded using a wearable eye-tracking device. After checking the distribution of the obtained outcomes and performing correlation analysis, a generalized linear model (GLM) was created to investigate the relationship between the number of glances at the smartphone and the participants' navigation performance and gaze behavior. To confirm the validity of the estimates for each GLM, a bootstrap sample of 2,000 iterations was used to evaluate the estimates.

Results: Correlation analysis showed that the number of stops and route errors were moderately and positively correlated with the number of times participants glanced at their smartphones ($P<0.05$). In the

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original GLM, the number of stops and route errors showed a significant positive association with the number of glances at the smartphone ($P<0.05$). This significant positive relationship persisted even after examining the bootstrap sample.

Conclusions: In healthy adults, frequent screen glances while walking with a navigation application may interfere with navigation performance. Establishing an alternative navigation system that does not rely on screen-checking may help pedestrians move more efficiently and safely.

Keywords: Application; gaze behavior; smartphone; navigation; real-world

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Introduction

The digitalization of modern society has brought several benefits, including easier information access, better communication, and increased efficiency in many aspects of people's lives. One way to benefit from digitalization is through smartphones and other mobile devices. Mobile devices are small and lightweight, and can be used as convenient information and communication terminals. Smartphones act as gateways for information at individuals' fingertips, making their daily lives more convenient. Recent surveys have shown that the number of smartphone users, ranging from young to older adults, is rapidly increasing (1,2). Although addiction due to excessive smartphone use is an important issue (3-5), many people use smartphones in their daily lives to make phone calls, exchange messages, check schedules, make purchases, pay bills, and access entertainment content; thus making their lives more convenient. Among the various functions of smartphones, route navigation plays an important role in supporting modern people's independent lives by providing directions to destinations and traffic information. Helping people reach their destinations, regardless of their gender or age, is essential for maintaining their independence and can affect their quality of life (6-8). Evidence shows improved personal health and economic status resulting from an increased range of activities (9). Particularly for older adults, maintaining mobility is important for successful aging (10), and maintaining the ability to walk may help extend healthy life expectancy (11). Therefore, using smartphones to support outdoor walking has several health benefits.

However, certain concerns about smartphone-based navigation exist. For example, looking at a screen while walking distracts the users and increases the risk of accidents. These behaviors are particularly prevalent among younger users, who use them more frequently while waiting at traffic lights and crossing crosswalks (12). A tendency toward smartphone addiction may exacerbate these risks by increasing walking inaccuracy and colliding with obstacles on the road (13). Additionally, smartphone use while walking is a growing safety concern in outdoor environments because of its potential to harm users and other road users (14). Although these reports primarily focused on the use of smartphone texting features and content viewing while walking, the aforementioned issues may be similar to smartphone navigation while walking. Gaze behavior during

Highlight box

Key findings

- Frequent glances at the smartphone during application-based pedestrian navigation were significantly associated with an increased number of stops and route errors.
- This association remained significant even after adjusting for age, gender, and bootstrap analyses.

What is known and what is new?

- Previous studies have shown that smartphone use while walking can distract users and impair their situational awareness, thereby increasing the risk of accidents and navigation errors.
- This study is the first to investigate the relationship between gaze behavior—specifically, the number of smartphone glances—and real-world pedestrian navigation performance using eye-tracking technology in a naturalistic outdoor setting.

What is the implication, and what should change now?

- These findings suggest that frequent glancing at a smartphone screen while navigating on foot can interfere with efficient and safe route-finding.
- Future navigation technologies should consider reducing the need for visual checks of the screen—for example, through haptic or auditory guidance systems—to promote safer pedestrian mobility in digitalized societies.

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smartphone use is of particular interest in the present study. Compared with normal walking, smartphone users may find that during smartphone navigation while walking their gaze is drawn to the screen, and their attention to the environment is dispersed. Previous studies have noted that in healthy participants during normal walking, gaze behavior characteristics vary with pedestrian position and direction, that is, a tendency to focus on approaching pedestrians, and the possibility of attention lateralization (15,16). Additionally, the gaze angle and walking speed were influenced by the complexity of the road surface; the more complex the road surface, the lower the gaze angle and the slower the walking speed (17). Another report showed a significant increase in the number of saccades in gaze behavior during navigation (18). Considering these findings, smartphone navigation while walking will likely require more eye movements to integrate navigational information with real-world information, which will affect navigation performance. In other words, frequent glances at smartphones do not necessarily positively affect navigational performance. Previous research has reported that saccade angle and fixation frequency decrease during smartphone navigation in older adults (19). However, this result only shows age-related changes in gaze behavior during smartphone navigation and does not investigate the effects of glancing at a smartphone on navigation performance in detail. To assess the navigation performance, measuring the number of route errors and stops is considered informative. Route errors reflect failures in spatial decision-making and the integration of map-based information with the real environment, which is a widely recognized marker of impaired wayfinding ability (20,21). The number of stops indicates interruptions and hesitation during a task, serving as an indicator of increased navigational difficulty and inefficiency (19,22,23).

Therefore, this study examined in detail the gaze behavior of adult smartphone users while using navigation applications, particularly glancing at a smartphone, and examined the effect of this behavior on their navigation performance. We hypothesized that frequent glancing at the smartphone during navigation would be negatively associated with navigation performance. The study results will help in understanding the risks involved in smartphone navigation while walking and provide criteria for the safe use of navigation aids in today's rapidly digitalized society. We present this article in accordance with the TREND reporting checklist (available at https://mhealth.amegroups.com/article/view/10.21037/mhealth-25-33/rc).

Methods

Sample size calculation

An A-priori sample size calculation was conducted using G*Power software (ver. 3.1.9.7) (24), assuming a linear regression model with an effect size (f^2) of 0.40, an α error probability of 0.05, a statistical power of 0.80, and three independent variables. This analysis indicated that a minimum sample size of 32 participants was required.

Participants

The participants involved healthy adults aged 20 years and older. They were recruited by posting an invitation to participate in the study on a bulletin board at Kagoshima University and a temporary agency for older adults in Kagoshima City, Japan. The inclusion criteria were: no history of neurological disease, no severe visual or hearing impairment, and no need for assistance in the general activities of daily living. Participants who needed assistance walking outdoors (including walking aids, such as canes or walkers) were excluded. All participants were paid a cooperation fee of 3,000 Japanese yen (approximately 20 USD). The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. This study was approved by the Ethics Review Committee of the Kagoshima University School of Medicine (No. 210282). Informed consent was obtained from all participants.

Route navigation task (RNT)

In earlier investigations (19,23), an RNT was employed to evaluate participants' navigational performance using a smartphone application in an unfamiliar outdoor environment. A designated outdoor course spanning 1,300 m was formulated, a locale that individuals perceived as akin to a "neighborhood outing" (25) (*Figure 1A*). Subsequently, the examiner directed the participants to traverse expeditiously from the initiation point to terminus, relying solely on the navigation application without soliciting guidance from external sources (examiners or local inhabitants). The examiner intervened only upon identifying a participant's navigational error, such as persisting along an incorrect path for more than 5 m, whereupon the participant was halted and instructed to scrutinize the smartphone screen attentively yet was not provided with corrective guidance. Throughout this process, the examiner trailed behind,

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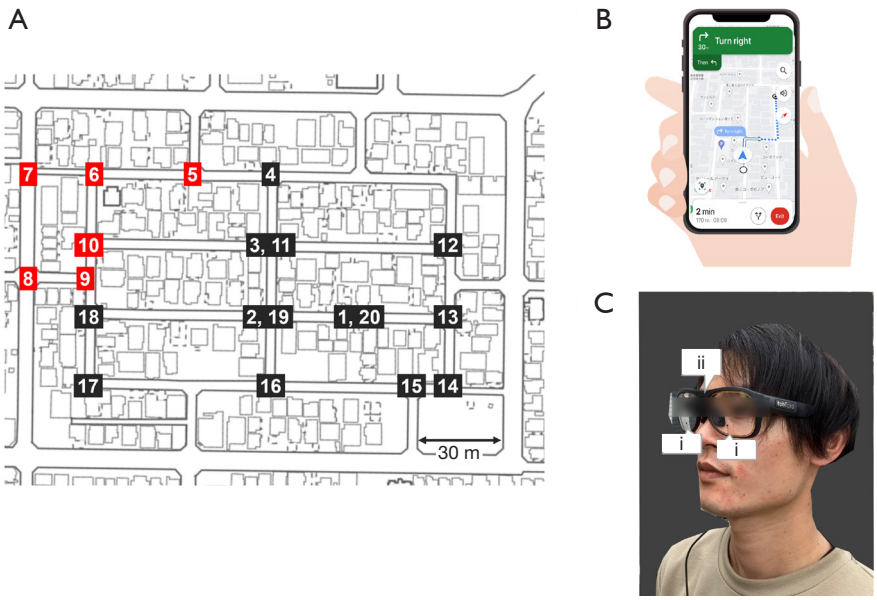


Figure 1 Equipment used for this study. (A) Implementation environment of RNT. The participants were asked to walk from #1 to #20 as quickly as possible using a navigation application. The participants' glances at their smartphones were counted while they walked the red-colored segment of the route from points #5 to #10. (B) A smartphone with the navigation application installed was held in either the left or right hand. The screen of the navigation application showed the current location and route to the destination, with the direction of travel facing upward. (C) An eye-tracking device in the form of glasses was worn at all times during the RNT, and gaze behavior during walking was recorded. Two cameras for eye tracking (I) and a high-definition scene camera in the center were used to record the forward viewpoint while walking using the navigation application (II). RNT, route navigation task.

ensuring the participants' safety and prompt responsiveness in the event of emergencies or unforeseen eventualities. The mobile devices used in this study were those routinely utilized by the participants and updated to the latest iterations of their respective operating systems (iOS or Android) and navigation applications (Google Maps; Google Inc., Mountain View, CA, USA). The navigation application settings were configured by the examiner to consistently display the upward direction during the RNT (*Figure 1B*). Besides the visual display, the application provided standard turn-by-turn audio prompts, similar to typical real-world applications. Both modalities were included to reproduce a naturalistic navigational aid environment. Parameters measured in the RNT included total time (s), number of stops, and route errors, all of which were documented by the accompanying examiner.

Evaluating gaze behavior during smartphone navigation while walking

To evaluate the participants' visual attention patterns during navigation, Tobii Pro Glasses 3 in conjunction with Tobii Pro Lab software (Tobii Technology, Danderyd, Sweden) were employed, as illustrated in *Figure 1C*. The Tobii Pro Glasses 3 system, designed to capture the binocular gaze direction, also provides supplementary data, including pupil size, three-dimensional orientation of each eye within a headset-fixed coordinate system, and gyroscope and accelerometer data indicating headset movement. The eye tracker operated at a sampling frequency of 100 Hz. The Tobii Pro Glasses series, recognized for its utility in diverse research contexts (26-28), has been extensively utilized as a head-mounted eye-tracking solution. Fixations and saccades were identified using the default I-VT filter within Tobii Pro Lab (velocity threshold: 30°, minimum fixation duration: 60 ms). After data collection, gaze information was extracted from the raw eye-tracking data using dedicated analysis software. The parameters extracted from the gaze information included entire fixation time, number of total fixations, number of total saccades, and degrees of whole saccades. Furthermore, the analysis of the concurrently recorded gaze behavior movie data facilitated

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Table 1 Summary statistics for each RNT outcome (n=33)

RNT outcomes	Mean	SD	Variance	Skewness	Shapiro-Wilk W	Shapiro-Wilk P
Total time in RNT (s)	824.9	113.3	12.84	0.92	0.95	0.10
Number of stops in RNT (n)	2.82	3.48	0.01	1.10	0.80	<0.001
Number of route errors in RNT (n)	0.67	1.22	0.001	3.02	0.59	<0.001
Whole fixation time in RNT (s)	408.1	183.8	33.78	–0.23	0.96	0.35
Number of total fixations in RNT (n)	1,286	483.2	233.5	–0.40	0.98	0.73
Number of total saccades in RNT (n)	885	332.2	110.4	–0.44	0.97	0.44
Degrees of whole saccades in RNT (degree)	7,171	3,650	13,300	0.38	0.97	0.56
Glancing at a smartphone (n)	22.73	12.26	0.15	1.15	0.89	0.002

RNT, route navigation task; SD, standard deviation.

the quantification of instances in which participants glanced at their smartphones during Segments #5 to #10 of the RNT course (*Figure 1A*). This segment was selected because it contained an appropriate balance of turns and straight walking, making it suitable for capturing representative gaze behaviors during the navigation task. Additionally, by focusing on the route's middle portion, we avoided potential start-up effects (when participants were still adapting to the task and equipment) and behaviors that may be related to anticipatory actions, such as increased goal-directed attention or urgency to complete the task, which could alter typical gaze patterns. In this study, a glance was operationally defined as a gaze directed at a smartphone screen that lasted for at least 300 ms, as determined by eye-tracking data. This threshold was selected to exclude very brief fixations or saccades and to capture purposeful screen-checking behavior.

Statistical analysis

First, we checked the distribution of all variables obtained using the RNT. The normality of the variables was checked by visual inspection using histograms and the Shapiro-Wilk test. We then calculated Spearman's correlation coefficients to examine the association between the number of times participants glanced at their smartphones and other variables in the RNT. Additionally, a generalized linear model (GLM) was created to examine how the number of times a participant glanced at their smartphone during the navigation walk was associated with each variable. The six dependent variables (DVs) of the RNT outcomes in the GLM were total time, number of stops, number of route errors, whole fixation time, number of total fixations, number of total saccades, degree of whole saccades, and glancing at the smartphone. In the GLM, the independent variable of interest was glancing at a smartphone, and age and gender were considered covariates. Because the study's primary aim was to explore potential associations rather than to make predictions, formal goodness-of-fit comparisons with alternative models were not conducted. Bootstrap resampling of 2,000 iterations was used to assess the validity of the estimates of the number of glances at a smartphone obtained by the GLM. Bootstrap method was used because the number of participants was insufficient to meet the study's objectives. The bootstrap method is considered more realistic for estimating the population from actual data (29). The significance level was set at less than 5% and R. ver. 4.3.2 was used for analysis and plotting. "car" package was used for bootstrapping.

Results

Table 1 presents the RNT results. Thirty-three participants (20–81 years, 66.7% female) were included in the analysis, and none were excluded. No unpredictable or adverse events occurred during the study. The number of RNT stops and route errors were assumed to be pseudo-Poisson distributed. Whereas, other variables were assumed to be normally distributed. *Figure 2* shows a scatterplot of the number of times participants looked at their smartphones while walking and other RNT outcomes. Correlation analysis showed that the number of stops and route errors were moderately and positively correlated ($P<0.05$) with the number of times participants looked at their smartphones. Conversely, associations with other variables were not significant.

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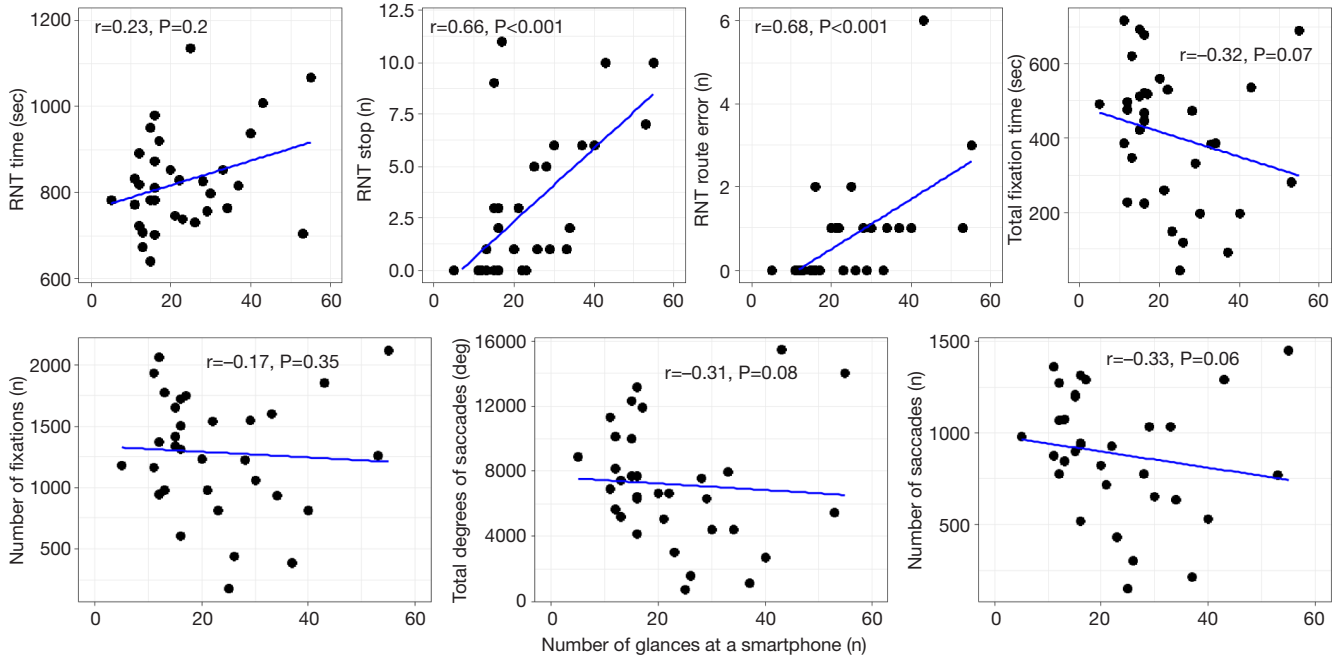


Figure 2 Correlation between the number of glances at a smartphone and outcomes of RNT. RNT, route navigation task.

Tables 2,3 summarize the original regression model and regression model with a bootstrapped sample of 2,000 iterations. Figure S1 shows the distribution of regression coefficient estimates obtained for the bootstrap sample. In both the original regression model and regression model with the bootstrapped sample, there was a significant positive association (P<0.05) among the number of stops, number of route errors, and number of times one looked at the smartphone. Specifically, as the number of times the participants looked at their smartphones increased, so did the number of stops and route errors.

Discussion

The number of times the participants looked at their smartphones while walking using a smartphone navigation application was found to be significantly positively associated with the number of stops and route errors, even after adjusting for age and gender. This result held for the bootstrapped samples. Conversely, there was no significant association with saccades or fixation gaze parameters and number of glances at a smartphone. Allocentric and egocentric strategies are commonly discussed in the literature as theoretical frameworks for understanding navigation (20,21,30,31). Although our study did not directly assess these strategies, they may provide a useful context for interpreting why frequent glancing at the smartphone is associated with increased stops and route errors. Specifically, difficulties integrating map-based (allocentric) and self-referential (egocentric) information could partly underlie the observed inefficiency. However, this remains speculative and should be investigated in future studies by using direct measures of navigational strategies. Additionally, because users used surrounding landmarks to obtain directions, the navigation application provided landmarks, such as the surrounding terrain, buildings, and intersections, which helped users determine their paths by referring to them. The Global Positioning System (GPS) function allowed the users to constantly monitor their progress and adjust the course of travel by comparing their position with the directions provided by the navigation application. Considering the participants checked their smartphones and stopped more often suggests that they may have had more difficulty integrating information on the screen map with the actual walking world. Thus, requiring more confirmation time. The relationship between glancing and increased route errors suggests that participants who had difficulty integrating information on the screen with the real world also had more difficulty selecting the correct route presented on the screen, although more glancing was required. Given that the number of glances was not related to the parameters of gaze behavior may be related to higher-

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Table 2 Associations between smartphone glances, navigation performance, and gaze parameters in the original regression models

Dependent variable	Independent variables	Estimate	SE	95% CI for estimate
Total time in RNT	(Intercept)	816.72***	39.51	739.28, 894.17
	Glancing at a smartphone	1.84	1.68	−1.45, 5.14
	Age	0.29	0.87	−1.42, 1.99
	Gender (male)	−142.89***	32.74	−207.06, −78.72
Number of stops in RNT	(Intercept)	0.11	0.53	−1.05, 1.07
	Glancing at a smartphone	0.05**	0.02	0.02, 0.08
	Age	−3.52e−3	0.01	−0.03, 0.02
	Gender (male)	−1.11*	0.54	−2.33, −0.16
Number of route errors in RNT	(Intercept)	−2.94**	0.94	−5.19, −1.38
	Glancing at a smartphone	0.05**	0.02	0.02, 0.08
	Age	0.02	0.02	−0.01, 0.06
	Gender (male)	−1.48*	0.69	−3.17, −0.32
Whole fixation time in RNT	(Intercept)	607.84***	73.82	463.16, 752.51
	Glancing at a smartphone	1.44	3.14	−4.72, 7.59
	Age	−4.18*	1.63	−7.37, −0.99
	Gender (male)	−85.88	61.16	−205.76, 34.00
Number of total fixations in RNT	(Intercept)	1,594.16***	209.63	1,183.30, 2,005.02
	Glancing at a smartphone	3.70	8.92	−13.78, 21.18
	Age	−5.88	4.62	−14.93, 3.17
	Gender (male)	−316.65	173.70	−657.09, 23.79
Number of total saccades in RNT	(Intercept)	1,188.23***	139.20	915.39, 1,461.06
	Glancing at a smartphone	3.01	5.92	−8.60, 14.61
	Age	−6.51*	3.07	−12.52, −0.50
	Gender (male)	−162.45	115.35	−388.53, 63.62
Degrees of whole saccades in RNT	(Intercept)	10,076.73***	1,494.95	7,146.69, 13,006.77
	Glancing at a smartphone	89.30	63.59	−35.34, 213.94
	Age	−92.01**	32.93	−156.55, −27.48
	Gender (male)	−1,347.96	1,238.72	−3,775.80, 1,079.88

*, P<0.05; **, P<0.01; ***, P<0.001. CI, confidence interval; RNT, route navigation task; SE, standard error.

level processing functions, such as cognition and awareness rather than sensory-level functions, including gaze behavior because real-world navigational gait is a cognitive function that requires multiple strategies and processing levels (32).

The regression analysis results indicated that the effects of gender (favoring men) on navigational performance and effects of aging on gaze behavior were also significant. Several studies have reported a male advantage in navigation (33-35), and the present study confirmed these results. Regarding gaze behavior during navigation, evidence shows the number of saccades and fixation time increasing with age (18). However, the present results show that the number and angle of saccades, and fixation time decrease with age. The reason for this difference is that, compared

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Table 3 Associations between smartphone glances, navigation performance, and gaze parameters in the 2,000-bootstrap regression models

Dependent variable	Independent variables	Median estimate	SE	Bias	95% CI for median estimate
Total time in RNT	(Intercept)	817.24*	32.02	1.08	756.23, 881.19
	Glancing at a smartphone	1.75	2.21	–0.09	–2.22, 6.51
	Age	0.29	0.96	–0.02	–1.67, 2.21
	Gender (male)	–139.25*	28.09	3.31	–204.12, –92.55
Number of stops in RNT	(Intercept)	0.08	1.09	–0.28	–2.55, 1.52
	Glancing at a smartphone	0.05*	0.02	2.68E–03	0.03, 0.09
	Age	–3.35E–03	0.02	2.61E–03	–0.04, 0.04
	Gender (male)	–1.19*	2.26	–0.47	–2.38, –0.28
Number of route errors in RNT	(Intercept)	–3.09*	1.24	–0.38	–5.07, –1.47
	Glancing at a smartphone	0.05*	0.03	0.01	0.01, 0.13
	Age	0.02	0.02	7.10E–04	–0.03, 0.07
	Gender (male)	–1.58	5.96	–2.34	–18.53, 0.51
Whole fixation time in RNT	(Intercept)	608.01*	60.2	2.71	492.65, 741.64
	Glancing at a smartphone	1.4	4.19	–0.27	–6.58, 10.43
	Age	–4.01*	1.7	–0.02	–8.4, –1.53
	Gender (male)	–80.6	59.62	4.71	–208.97, 29.64
Number of total fixations in RNT	(Intercept)	1,609.54*	174.29	25.11	1,287.02, 1,972.01
	Glancing at a smartphone	3.21	12.25	–1.62	–22.41, 29.04
	Age	–5.49	4.51	–0.07	–17.09, 1.04
	Gender (male)	–318.01	165.9	6.32	–615.07, 45.32
Number of total saccades in RNT	(Intercept)	1,196.43*	111.59	9.46	947.17, 1,402.75
	Glancing at a smartphone	2.87	8.19	–0.79	–13.62, 20.11
	Age	–6.24*	3.07	–0.06	–15.59, –1.93
	Gender (male)	–155.43	107.57	7.27	–371.02, 49.97
Degrees of whole saccades in RNT	(Intercept)	10,096.71*	1,216.51	10.21	7,541.43, 12,428.75
	Glancing at a smartphone	95.83	87.74	–2.81	–114.45, 240.85
	Age	–90.37*	27.68	–0.41	–161.3, –46.93
	Gender (male)	–1,304.61	1,120.25	54.88	–3,398.4, 960.93

*, P<0.05. If the bootstrapped 95% confidence interval does not cross 0, P<0.05 is indicated. CI, confidence interval; RNT, route navigation task; SE, standard error.

to the normal navigation task, older adults using the application paid more attention to taking in the navigation information on the smartphone than to perceiving the entire environment in which they were walking. Thus, their eye movements were limited to a narrow range between the smartphone screen and direction in which they were walking such that the smartphone screen was always in view. Particularly, younger adults may have entered more navigational information with fewer inputs than older adults and may have had a wider view of their surroundings while walking. Therefore, these results highlight that looking at a smartphone while walking not only degrades navigation

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performance, but may also narrow the field of view in older adults. Currently, new navigation systems, such as devices using haptic technology (36,37) and navigation-assistive technologies using augmented reality (AR) technology (38,39) are being developed. These technologies are likely to help users safely navigate to their destinations in a more intuitive manner without repeatedly looking at the smartphone screen. Future research should closely compare the effectiveness of these navigation-assisted technologies with that of navigation through smartphone applications to contribute to safer and more efficient mobility.

This study has certain limitations that should be considered when interpreting the results. First, the study results are based on a limited sample size of healthy adults and should be generalized with caution. However, the results from the 2,000 bootstrap samples support the present results. Second, participants were unfamiliar with the route. Given that egocentric strategies tend to be preferentially used, especially for familiar routes (30), the results when using routes familiar to the participants could be different. Third, the number of smartphone glances was quantified only on a subset of routes (between points #5 and #10). This segment was chosen because it contained both turns and straight paths. Thus, representing typical gaze patterns, and because it avoided starting- and ending-point effects. Nevertheless, gaze behavior may vary in other route segments, and future studies should aim to capture glancing behavior throughout the route. Finally, we could not determine the causal direction of the observed association between smartphone glances and navigational performance. Possibly, frequent glancing contributed to poorer performance. However, it is equally plausible that participants glanced more often as a compensatory behavior when experiencing difficulty or after making errors. As our data did not allow for a detailed temporal sequence analysis, future research should examine the order of events to clarify whether smartphone glancing primarily impairs navigation or reflects attempts to recover from uncertainty. Despite these limitations, our study is informative as it focuses on the functionality of navigation applications in the current era of increasingly popular smartphones and their impact on navigation performance in terms of participants' eye-movement behavior.

Conclusions

This study examined in detail the performance of 33 young to older adults in navigational walking using a smartphone navigation application based on the participants' gaze behavior. The results showed that those who glanced at a smartphone more often were more likely to exhibit poor navigation performance. This suggests that participants may have difficulty integrating on-screen information with real-world information while walking, highlighting the need for alternatives to achieve safer and more efficient navigation.

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Footnote

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